

Problem H. Two-gram

Time Limit 1000 ms

Mem Limit 262144 kB

Two-gram is an ordered pair (i.e. string of length two) of capital Latin letters. For example, "AZ", "AA", "ZA" — three distinct two-grams.

You are given a string s consisting of n capital Latin letters. Your task is to find **any** two-gram contained in the given string **as a substring** (i.e. two consecutive characters of the string) maximal number of times. For example, for string $s = \text{"BBAABBB"}$ the answer is two-gram "BB", which contained in s three times. In other words, find any most frequent two-gram.

Note that occurrences of the two-gram can overlap with each other.

Input

The first line of the input contains integer number n ($2 \leq n \leq 100$) — the length of string s . The second line of the input contains the string s consisting of n capital Latin letters.

Output

Print the only line containing exactly two capital Latin letters — **any** two-gram contained in the given string s **as a substring** (i.e. two consecutive characters of the string) maximal number of times.

Examples

Input	Output
7 ABACABA	AB

Input	Output
5 ZZZAA	ZZ

Note

In the first example "BA" is also valid answer.

In the second example the only two-gram "ZZ" can be printed because it contained in the string "ZZZAA" two times.